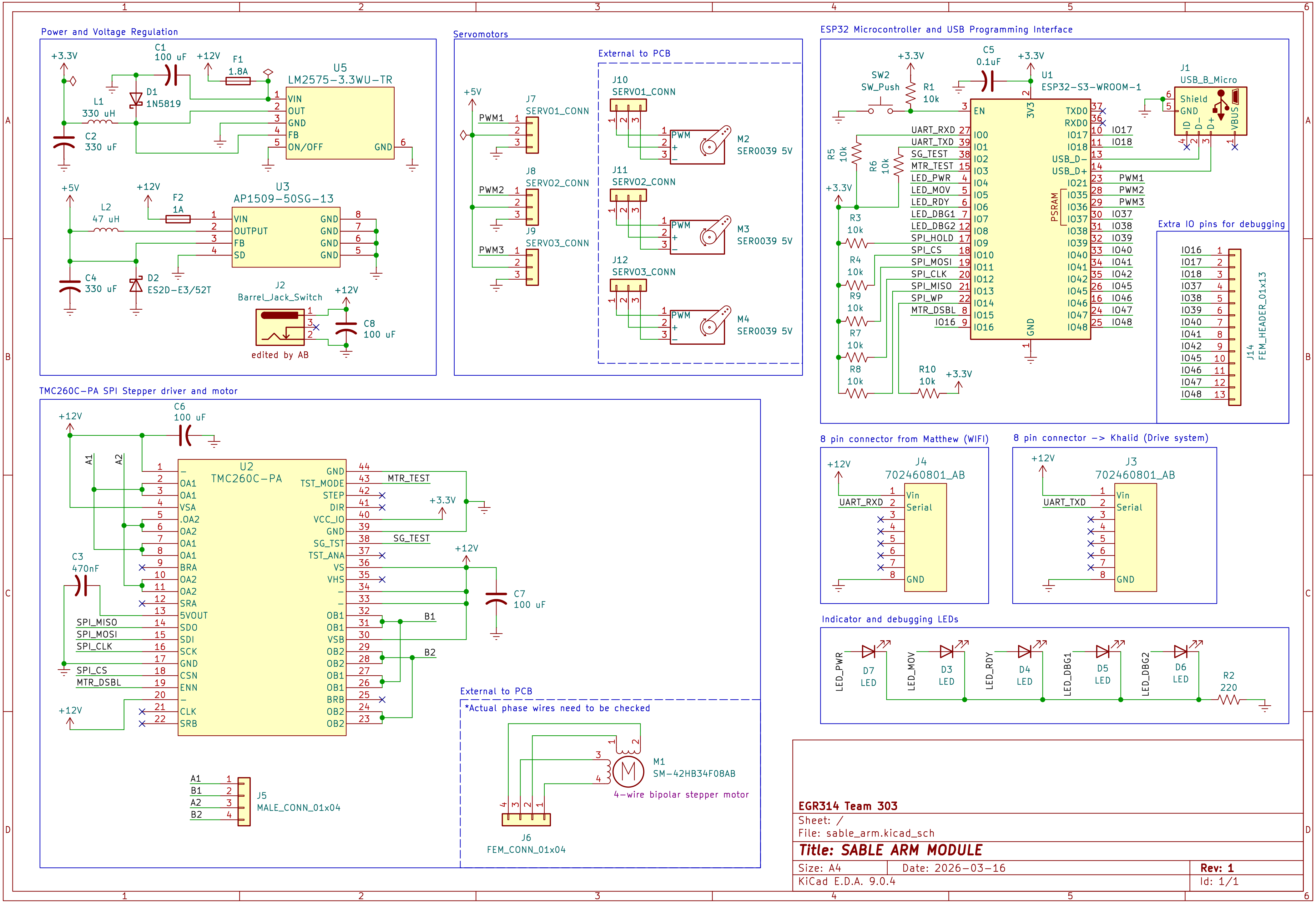

Power and Voltage Regulation
+3.3V
C1
100 uF
+12V
F1
1.8A
U5
LM2575-3.3WU-TR
VIN
OUT
GND
FB
ON/OFF
GND
L1
330 uH
D1
1N5819
C2
330 uF
+5V
+12V
F2
1A
U3
AP1509-50SG-13
VIN
OUTPUT
FB
SD
GND
L2
47 uH
C4
330 uF
D2
ES2D-E3/52T
J2
Barrel_Jack_Switch
+12V
C8
100 uF
edited by AB
Servomotors
External to PCB
+5V
PWM1
J7
SERVO1_CONN
PWM2
J8
SERVO2_CONN
PWM3
J9
SERVO3_CONN
J10
SERVO1_CONN
M2
SER0039 5V
J11
SERVO2_CONN
M3
SER0039 5V
J12
SERVO3_CONN
M4
SER0039 5V
PWM
ESP32 Microcontroller and USB Programming Interface
+3.3V
C5
0.1uF
+3.3V
U1
ESP32-S3-WROOM-1
J1
USB_B_Micro
SW2
SW_Push
R1
10k
EN
3V3
TXD0
RXD0
UART_RXD
UART_TXD
SG_TEST
MTR_TEST
LED_PWR
LED_MOV
LED_RDY
LED_DBG1
LED_DBG2
SPI_HOLD
SPI_CS
SPI_MOSI
SPI_CLK
SPI_MISO
SPI_WP
MTR_DSBL
IO16
IO17
IO18
USB_D-
USB_D+
PWM1
PWM2
PWM3
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
GND
Shield
GND
ID
D-
D+
VBUS
R5
10k
R6
10k
R3
10k
R4
10k
R9
10k
R7
10k
R8
10k
R10
10k
+3.3V
Extra IO pins for debugging
J14
FEM_HEADER_01x13
TMC260C-PA SPI Stepper driver and motor
+12V
C6
100 uF
A1
A2
U2
TMC260C-PA
OA1
VSA
OA2
BRA
SRA
5VOUT
SDO
SDI
SCK
GND
CSN
ENN
CLK
SRB
GND
TST_MODE
STEP
DIR
VCC_IO
SG_TST
TST_ANA
VS
VHS
OB1
VSB
OB2
BRB
MTR_TEST
+3.3V
SG_TEST
+12V
C7
100 uF
B1
B2
C3
470nF
SPI_MISO
SPI_MOSI
SPI_CLK
SPI_CS
MTR_DSBL
+12V
J5
MALE_CONN_01x04
A1
B1
A2
B2
External to PCB
*Actual phase wires need to be checked
M1
SM-42HB34F08AB
4-wire bipolar stepper motor
J6
FEM_CONN_01x04
8 pin connector from Matthew (WIFI)
+12V
J4
702460801_AB
UART_RXD
Vin
Serial
GND
8 pin connector -> Khalid (Drive system)
+12V
J3
702460801_AB
UART_TXD
Vin
Serial
GND
Indicator and debugging LEDs
LED_PWR
D7
LED
LED_MOV
D3
LED
LED_RDY
D4
LED
LED_DBG1
D5
LED
LED_DBG2
D6
LED
R2
220
EGR314 Team 303
Sheet: /
File: sable_arm.kicad_sch
Title: SABLE ARM MODULE
Size: A4
Date: 2026-03-16
KiCad E.D.A. 9.0.4
Rev: 1
Id: 1/1